

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 20
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data handled in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported dataset using Pandas.
- **Initial Exploration:** Used `df.info()` and `describe()`.
- **Missing Data Handling:** Filled null review ratings using category median.
- **Column Standardization:** Converted names to snake_case.
- **Feature Engineering:**
 - Created `age_group`
 - Created `purchase_frequency`

- **Duplicate Removal:** Removed redundant promo code column.

	customer_id	age	purchase_amount	review_rating	previous_purchases	purchase_frequency
count	3900.000000	3900.000000	3900.000000	3900.000000	3900.000000	3900.000000
mean	1950.500000	44.087436	59.764359	3.750051	25.351538	89.133077
std	1125.977353	15.215450	23.685392	0.713590	14.447125	119.037566
min	1.000000	18.000000	20.000000	2.500000	1.000000	7.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000	14.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000	30.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000	90.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000	365.000000

Figure 1: Dataset Preview Screenshot

4. Data Analysis using SQL

We performed structured SQL analysis to answer business questions:

1. Revenue by Gender
2. High-Spending Discount Users
3. Top 5 Products by Rating
4. Shipping Type Comparison
5. Subscribers vs Non-Subscribers
6. Discount-Dependent Products
7. Customer Segmentation
8. Top 3 Products per Category
9. Repeat Buyers & Subscriptions
10. Revenue by Age Group

1. Revenue by Gender

Compared total revenue generated by male and female customers.

	Gender	revenue
0	Female	75191
1	Male	157890

Figure 2: Revenue by Gender

2. High-Spending Discount Users

Identified customers who used discounts but still spent more than the average purchase amount.

	Customer ID	Purchase Amount (USD)
0	2	64
1	3	73
2	4	90
3	7	85
4	9	97
...	...	...
834	1667	64
835	1671	73
836	1673	73
837	1674	62
838	1676	90

839 rows × 2 columns

Figure 3: Discount Users

3. Top Rated Products

Found products with highest average review ratings.

	Item Purchased	Average_Product_Rating
0	Gloves	3.86
1	Sandals	3.84
2	Boots	3.82
3	Hat	3.80
4	Skirt	3.79

Figure 4: Product Ratings

4. Shipping Type Comparison

Compared average purchase amounts between Standard and Express shipping.

	Shipping Type	avg_purchase
0	Express	60.48
1	Standard	58.46

Figure 5: Shipping Type

5. Subscribers vs Non-Subscribers

Compared total customers, average spend, and revenue.

	Subscription Status	total_customers	avg_spend	total_revenue
0	No	2847	59.87	170436.0
1	Yes	1053	59.49	62645.0

Figure 6: Subscription Analysis

6. Customer Segmentation

Customers were grouped into:

- New
- Returning
- Loyal

	customer_segment	Number of Customers
0	Loyal	3116
1	New	83
2	Returning	701

Figure 7: Customer Segmentation

5. Dashboard in Power BI

Finally, an interactive dashboard was developed in Power BI to present insights visually.
Dashboard file: `customer_shopping_behavior.pbix`



Figure 8: Power BI Dashboard Screenshot

Dashboard includes:

- KPI Cards
- Revenue by Category
- Revenue by Gender
- Subscription Analysis
- Customer Segmentation
- Interactive Filters / Slicers

6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers.
- **Review Discount Policy** – Balance promotions with profit margins.
- **Product Positioning** – Highlight top-rated and best-selling products.
- **Targeted Marketing** – Focus on high-revenue age groups.

7. Conclusion

This project successfully transformed raw retail data into actionable business insights using Python, SQL, and Power BI. It demonstrates skills in analytics, data cleaning, SQL querying, dashboarding, and business intelligence.